



JURONG JUNIOR COLLEGE
2018 JC 2 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CLASS

18S24

EXAM INDEX

CHEMISTRY

8873/01

Higher 1

Paper 1 Multiple Choice

13 September 2018

1 hour

Candidates answer on separate paper.

Additional Materials: Multiple Choice Answer Sheet
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, class and exam index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** or **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

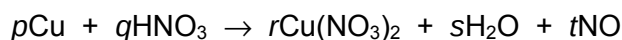
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

- 1 When copper reacts with a 50% solution of nitric acid, nitrogen monoxide is evolved and a blue solution results.

The balanced equation for this reaction is shown.



What are the values of the integers p , q , r , s and t ?

	p	q	r	s	t
A	1	4	1	2	2
B	2	6	2	3	2
C	2	8	2	4	4
D	3	8	3	4	2

- 2 A sample of silicon contains three naturally occurring isotopes, ^{28}Si , ^{29}Si and ^{30}Si . The sample is made up of 92.23% ^{28}Si and the relative atomic mass of silicon in this sample is 28.10.

What is the percentage of the isotope ^{29}Si in the sample?

- A** 2.23% **B** 3.89% **C** 5.54% **D** 7.77%

- 3 The ^{68}Ge isotope is medically useful because it undergoes a natural radioactive process to give an isotope of a different element, ^{68}X , which can be used to detect tumours. This transformation of ^{68}Ge occurs when an electron enters the nucleus and changes a proton into a neutron.

Which statement about the composition of an atom of ^{68}X is correct?

- A** It has 4 electrons in its outer p orbitals.
B It has 13 electrons in its outer shell.
C It has 37 neutrons.
D Its proton number is 32.

- 4 Drinking water may contain dissolved calcium hydrogencarbonate, $\text{Ca}(\text{HCO}_3)_2$.

How many electrons are present in a hydrogencarbonate anion?

- A** 30 **B** 31
C 32 **D** 33

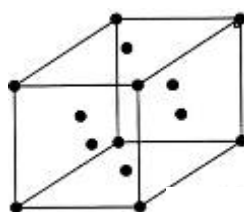
5 In which structure are there three atoms bonded together in a straight line?

- A Poly(ethene), $\left[\text{CH}_2 - \text{CH}_2 \right]_n$
- B Propane, C_3H_8
- C Silicon tetrachloride, SiCl_4
- D Sulfur hexafluoride, SF_6

6 At room temperature and pressure, H_2O is a liquid and H_2S is a gas.

What is the reason for this difference?

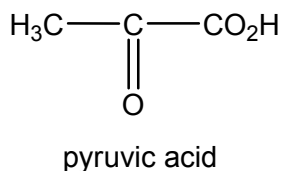
- A O has higher first and second ionisation energies than S.
- B The covalent bond between O and H is stronger than the covalent bond between S and H.
- C There is significant hydrogen bonding between H_2O molecules but not between H_2S molecules.
- D The instantaneous dipole-induced dipole forces between H_2O molecules are stronger than the instantaneous dipole-induced dipole forces between H_2S molecules.
- 7 Copper and iodine are both shiny crystalline solids. The crystal structures of copper and iodine are both face-centred cubic. The diagram shows the arrangement of the particles in this type of crystal lattice.



What are the particles present in each lattice?

	Copper	iodine
A	atoms	anions
B	atoms	atoms
C	cations	atoms
D	cations	molecules

- 8 The diagram shows the structure of pyruvic acid.



Which set of bond angles in pyruvic acid would be predicted by Valence Shell Electron Pair Repulsion theory?

- A 105°, 109°, 120°
 B 105°, 120° only
 C 109°, 120°, 180°
 D 109°, 120° only
- 9 Which row describes and explains the trend in electronegativity from fluorine to iodine?

	trend in electronegativity	explanation
A	decreases	the bonding pair of electrons becomes further away from the nucleus and so is less attracted to it
B	decreases	ionisation energy decreases down the group because the outer electron is strongly attracted to the nucleus
C	increases	the bonding pair of electrons becomes further away from the nucleus and so is less attracted to it
D	increases	ionisation energy decreases down the group because the outer electron is strongly attracted to the nucleus

- 10 Which types of bond are broken and formed in the addition polymerisation of alkenes?

	Type of bond broken	Type of bond formed
A	π only	σ only
B	π only	σ and π
C	σ and π	σ only
D	σ and π	σ and π

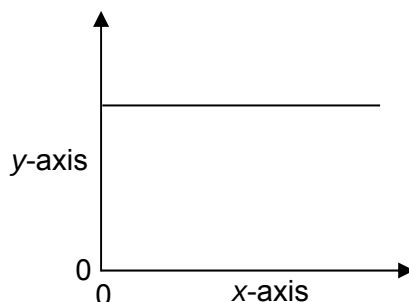
- 11** All the reactants and products of an endothermic reaction are gaseous.

Which statements about this reaction is correct?

- 1 On the reaction pathway diagram, the energy of the products are lower than that of the reactants.
- 2 There is a net transfer of heat energy from the surroundings to the reacting system.
- 3 The total bond energy of the reactants is more than the total bond energy of the products.

- A** 1, 2 and 3
B 1 and 2 only
C 2 and 3 only
D 3 only

- 12** The kinetics of the zero order reaction $U \rightarrow V$ were investigated under different conditions. The table shows pairs of quantities that were plotted as graphs. Which pairs gave the following graph?



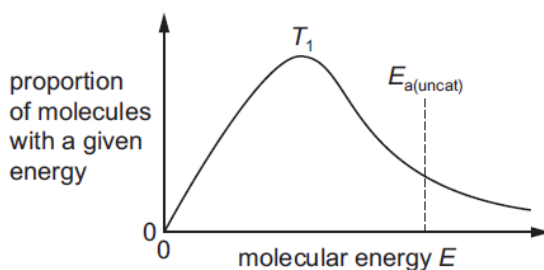
	y-axis	x-axis
1	rate	time
2	rate constant	time
3	rate constant	temperature

- A** 1, 2 and 3
B 1 and 2 only
C 2 and 3 only
D 1 and 3 only

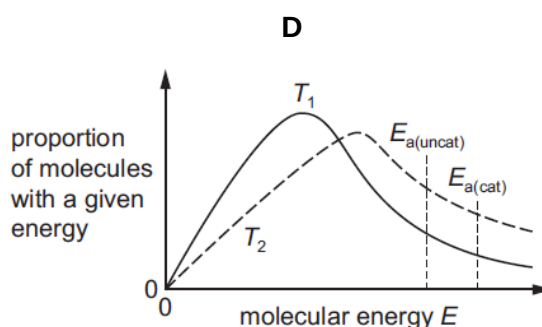
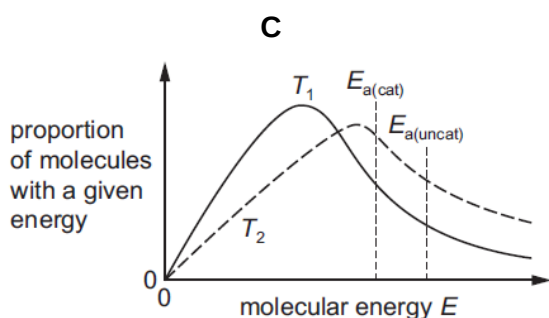
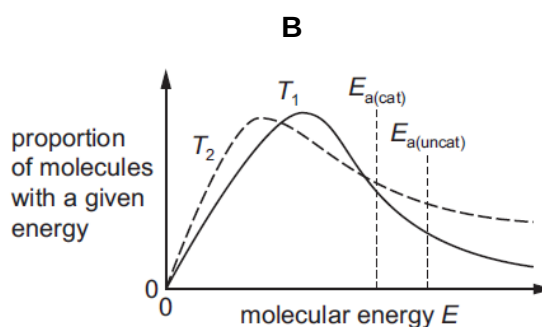
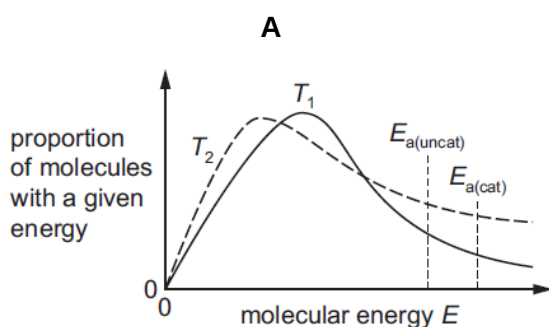
13 Which statement about the effect of a catalyst on a reversible reaction is correct?

- A The activation energy of the forward reaction stays the same.
- B The composition of the equilibrium mixture stays the same.
- C The rate of the backward reaction stays the same.
- D The value of the equilibrium constant changes.

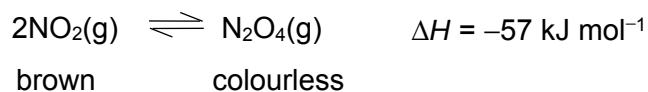
14 The diagram shows the distribution of molecular energies in a sample of gas at a temperature T_1 . The activation energy for an uncatalysed reaction of this gas, $E_{a(\text{uncat})}$, is shown.



Which diagram correctly shows the new distribution and new activation energy, $E_{a(\text{cat})}$, when the temperature is increased to T_2 , and a catalyst is used that increases the rate of reaction?



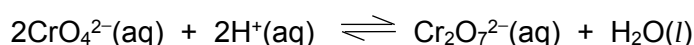
- 15** Nitrogen dioxide, NO_2 , is a brown gas.
 Dinitrogen tetroxide, N_2O_4 , is a colourless gas.
 An equilibrium is established between NO_2 and N_2O_4 in a closed vessel.



Which row describes the effects of changing conditions on the colour of an equilibrium mixture of NO_2 and N_2O_4 ?

	Increasing the pressure	Increasing the temperature
A	Colour becomes darker	Colour becomes darker
B	Colour becomes darker	Colour becomes lighter
C	Colour becomes lighter	Colour becomes darker
D	Colour becomes lighter	Colour becomes lighter

- 16** The following equilibrium is an exothermic reaction in the forward direction.



What happens when the concentration of $\text{CrO}_4^{2-}(\text{aq})$ ions **increases** and the temperature **decreases**?

- 1 The concentration of $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$ increases.
- 2 The equilibrium constant for the forward reaction increases.
- 3 The rate constants for the forward and reverse reactions increase equally.

- A** 1, 2 and 3
B 1 and 2 only
C 2 and 3 only
D 1 only

- 17** The compound $(\text{CH}_3)_3\text{NAlCl}_3$ has a simple molecular structure.

Which statement about $(\text{CH}_3)_3\text{NAlCl}_3$ is correct?

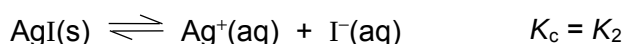
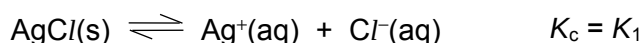
- A** $(\text{CH}_3)_3\text{NAlCl}_3$ molecules attract each other by hydrogen bonds.
B The Al atom has an incomplete valence shell of electrons.
C The bonds around the Al atom are planar.
D The molecules contain coordinate and covalent bonding.

- 18 A solution was made by mixing 0.002 mol of $\text{H}_2\text{SO}_4(\text{aq})$ and 0.005 mol of $\text{KOH}(\text{aq})$. Water was added until the volume of the solution was 1 dm^3 .

What is the pH of the solution?

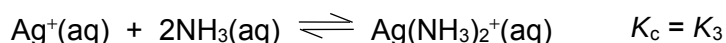
- A 11.0 B 11.5 C 11.7 D 12.0

- 19 Silver chloride and silver iodide form equilibria when added to water.



Each equilibrium position lies well to the **left**.

Silver iodide will not dissolve in aqueous ammonia. Silver chloride will dissolve in aqueous ammonia. In both cases, adding aqueous ammonia will form the following equilibrium.

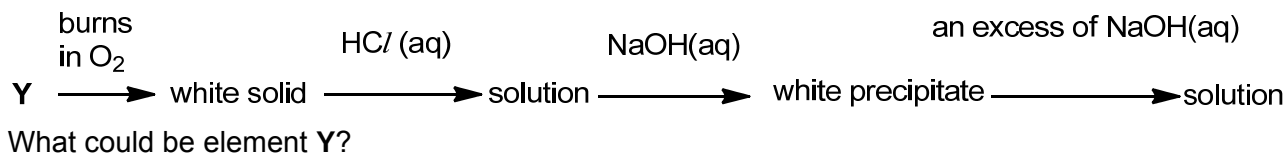


The position of this equilibrium lies to the **right**.

What is the order of magnitude for these three equilibrium constants?

- A $K_2 > K_3 > K_1$
 B $K_2 > K_1 > K_3$
 C $K_3 > K_1 > K_2$
 D $K_3 > K_2 > K_1$

- 20 An element Y reacts according to the following sequence.



- A Al B Ca C Mg D P

- 21** X_2 , Y_2 , Z_2 represent different halogens. The table shows the results of nine experiments in which aqueous solutions of X_2 , Y_2 and Z_2 were separately added to separate aqueous solutions containing X^- , Y^- and Z^- ions.

	$X^-(aq)$	$Y^-(aq)$	$Z^-(aq)$
$X_2(aq)$	No reaction	No reaction	No reaction
$Y_2(aq)$	X_2 formed	No reaction	Z_2 formed
$Z_2(aq)$	X_2 formed	No reaction	No reaction

Which row of the following table contains the ions X^- , Y^- and Z^- in order of their decreasing strength as reducing agents?

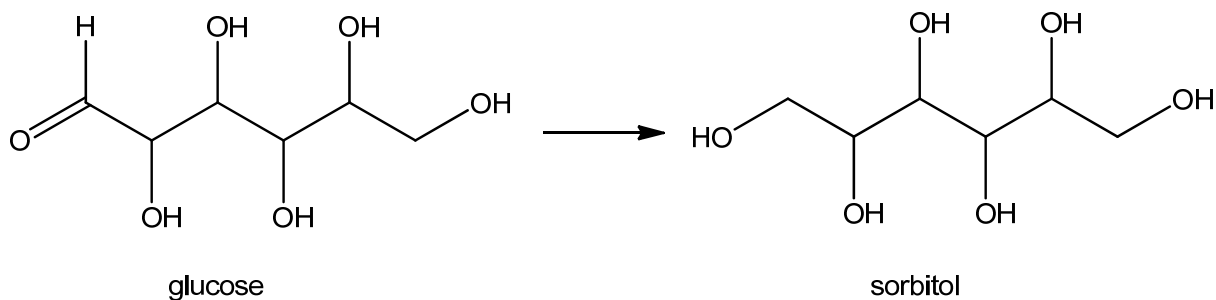
	strongest	→	weakest
A	X^-	Y^-	Z^-
B	X^-	Z^-	Y^-
C	Y^-	Z^-	X^-
D	Z^-	X^-	Y^-

- 22** 1-bromopropane undergoes a variety of reactions.

Which row is correct?

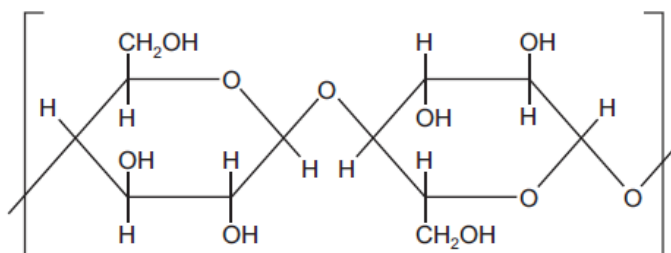
	Reagent added to 1-bromopropane	Products include
A	Hot NaOH(aq)	Propene
B	Cold NaOH(aq)	Prop-2-ene
C	Hot NaOH in ethanol	Propene
D	Hot NaOH in ethanol	Propan-1-ol

- 23 Glucose can be used to prepare sorbitol, a compound used as a sugar substitute.



Which reagent may be used for this conversion?

- A Acidified potassium manganate(VII)
 - B Acidified potassium dichromate(VI)
 - C Sodium hydroxide
 - D Sodium borohydride
- 24 A cathedral in New Zealand has been constructed from cardboard. Cardboard contains polymer molecules. Part of one such polymer molecule is shown below.



Which statements about this polymer are correct?

- 1 The polymer molecules can form hydrogen bonds with each other.
 - 2 The polymer can form intermolecular forces with water molecules.
 - 3 The polymer will not burn easily because it is a secondary alcohol.
- A 1, 2 and 3
 - B 1 and 2 only
 - C 2 and 3 only
 - D 1 only

- 25 Graphene, graphite and the fullerene C₆₀ are allotropes of carbon.

Which statements are correct for all three of these allotropes of carbon?

- 1 Delocalised electrons are present in the structure.
- 2 All bond angles are 120°.
- 3 It has a giant molecular crystalline lattice structure.

- A 1, 2 and 3
 B 1 and 2 only
 C 2 and 3 only
 D 1 only

- 26 PVC, $\left[\text{CH}_2 - \text{CHCl} \right]_n$ is difficult to dispose of.

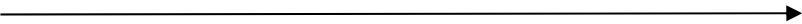
Two possible methods are burying it in landfill sites and disposal by combustion.

Which row of the table is correct?

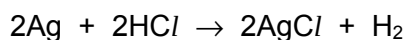
	Rate of biodegradation of PVC in landfill sites	Gases produced when PVC combusts
A	Fast	CO ₂ , H ₂ O, HCl
B	Fast	CO ₂ , H ₂ O, Cl ₂
C	Slow	CO ₂ , H ₂ O, Cl ₂
D	Slow	CO ₂ , H ₂ O, HCl

- 27 The gases butane, 2-methylpropane and propane are kept in a sealed container at atmospheric pressure and room temperature.

Which row represents the order in which these gases will liquefy as the pressure in the container is gradually increased?

	Increasing pressure		
			
A	butane	2-methylpropane	propane
B	2-methylpropane	butane	propane
C	propane	butane	2-methylpropane
D	propane	2-methylpropane	butane

- 28 Silver foil does not react when added to a beaker of dilute hydrochloric acid. However, silver nanoparticles do react with dilute hydrochloric acid.



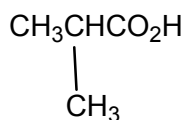
Which statement could explain why silver nanoparticles are more reactive than silver foil?

- A They act as catalysts for the reaction.
 B They contain a much greater concentration of silver.
 C They do not have an unreactive oxide layer.
 D They have a very high surface area to volume ratio.
- 29 Which molecular formula could represent molecules that exhibit *cis-trans* isomerism?

1	2	3	4
$\text{C}_3\text{H}_6\text{BrI}$	$\text{C}_3\text{H}_5\text{I}$	$\text{C}_3\text{H}_4\text{I}_2$	$\text{C}_3\text{H}_4\text{BrI}$

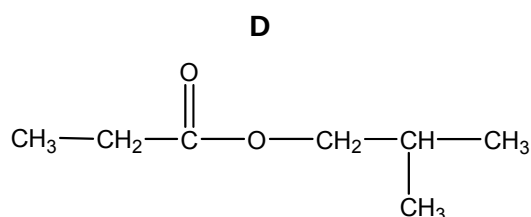
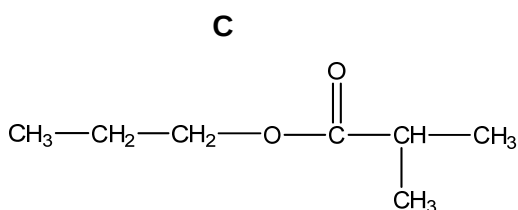
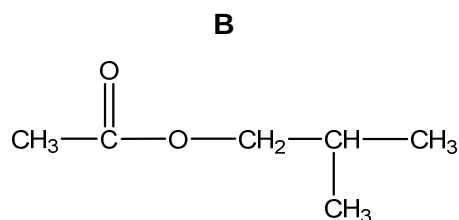
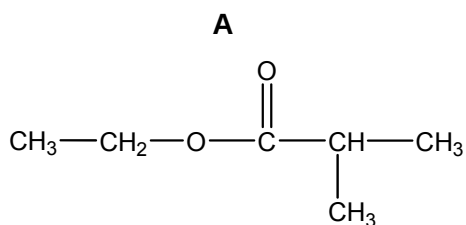
- A 1, 2 and 3 B 2, 3 and 4 C 2 and 4 only D 3 and 4 only

- 30 Ethyl propanoate is refluxed with aqueous sodium hydroxide. The alcohol produced is then reacted with methylpropanoic acid to make a second ester.



methylpropanoic acid

What is the structural formula of this second ester?



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